

REHATBIR SINGH

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EDUCATION

University of Wisconsin-Madison

BS Computer Science & Data Science (Dean's List) *GPA: 4.0 / 4.0*

Madison, WI, USA

Sep 2024 - May 2027

EXPERIENCE

Flux

Co-founder & Lead Software Engineer

Madison, WI, USA

Feb 2025 - Present

- Led product development for an app connecting student entrepreneurs by complementary skills, growing to 70+ MAU through rapid iteration and user feedback
- Made key decisions including adopting card-based profiles to enable future features and switching from Flutter to SwiftUI after validating iOS-dominant market to improve UX and development velocity
- Designed and implemented a semantic matching system using text embeddings, enabling flexible representations of users' skills and experiences

Al Masaood LLC

Software Developer Intern

Abu Dhabi, United Arab Emirates

May 2025 - Aug 2025

- Built and launched a React/SPFx timesheet tracker for 30+ technicians, saving 1,500+ sheets weekly through SAP-integrated SharePoint
- Assisted development of a legal chatbot using FastAPI, React, and RAG, saving the legal team 8 hrs/wk
- Partnered with the insurance department on an app for vehicle inventory and insurance, handling data modeling and SharePoint API integration

Data Science for Sustainable Development @ UW-Madison

Student Developer

Madison, WI, USA

Sep 2024 - Dec 2024

- Built a statistics dashboard for the open-source Red-CORAL crime visualization project, analyzing 1,200+ data points using React and Firebase
- Designed a modular filtering system using React Reducers, enabling flexible data slicing and real-time geospatial visualization

PROJECTS

FreeVoIP Rust, TypeScript, Tauri, React.js, Iroh

<https://rehatsingh.cv/freevoip>

Developed a decentralized peer-to-peer video calling app using Rust, Tauri, and Iroh, eliminating centralized services. Designed secure contact sharing, debugged a custom media pipeline, and resolved high-throughput Rust-JS data transfer limitations.

Path Tracing Renderer C, OpenGL, GLSL

https://github.com/MysteryCoder456/path_tracer_c

Developed a Monte Carlo path tracing renderer using C and OpenGL, implementing BSDF-based light transport and adapting a recursive CPU-bound algorithm to an iterative GLSL implementation to support multi-bounce light accumulation on the GPU.

Blitz Board Python, JavaScript, Flask, PostgreSQL, Redis

<https://blitz.rehatsingh.cv>

Built a real-time multiplayer typing game using WebSockets and Redis-backed shared state to synchronize game lobbies across multiple Flask servers. Identified and debugged client desynchronization issues under concurrent updates. Containerized the system using Docker and self-hosted it using Nginx.

AWARDS

Honorable Mention in MadData Hackathon

dotData @ UW-Madison

Recognized for building & presenting Wavelength, an app that analyzes Spotify libraries to connect students with similar music tastes using data science techniques and social discovery algorithms.

Feb 2025

Dean's List

University of Wisconsin-Madison

Recognized on the Dean's List at the University of Wisconsin-Madison for achieving a 4.0 GPA, signifying academic excellence.

Fall 2024 - Fall 2025

SKILLS

Frontend: JavaScript/TypeScript, React.js, Tailwind, Swift, SwiftUI
Backend: Python, Flask, Django, Rust, Axum, Node.js, Next.js
Infrastructure: Docker, Nginx, Linux, PostgreSQL, MySQL, Firebase